AI-Powered Appliance Pricing Intelligence Platform - Master Blueprint V2

Vision

Build a production-style ML product that helps consumers determine whether an appliance is fairly priced and discover better-value alternatives. The regression model is only one subsystem inside a pricing intelligence platform.

Business Problem

Consumers compare products mainly on brand and price. They often overpay because they cannot objectively determine whether a product's specifications justify its price. The platform estimates fair market value, highlights price premiums, calculates value-for-money scores, and recommends alternatives.

Core User Stories

1. Is this appliance worth its listed price? 2. Is the brand charging a premium? 3. Which specifications are driving the price? 4. Are there similar products with better value? 5. What should be the fair price for this product?

Dataset Design

Source: Smartprix. Required columns: Product Name, Brand, Category, Price, Rating, Review Count, Energy Rating, Capacity, Inverter Flag, Warranty, Technical Specifications. Target: Price. Features: All specifications except price.

Project Folder Structure

project/ data/raw data/processed notebooks src/scraping src/preprocessing src/features src/models src/inference src/api frontend models reports tests

Environment Setup

Python 3.11 pandas numpy scikit-learn xgboost matplotlib plotly shap beautifulsoup4 requests fastapi uvicorn joblib postgresql nextjs

Phase 1 - Data Collection

Scrape appliance categories. Store raw data. Create reusable scraping pipeline. Maintain structured schemas.

Phase 2 - Data Engineering

Handle missing values. Remove duplicates. Normalize units. Standardize brands. Parse specifications into structured columns.

Phase 3 - EDA

Price distributions. Brand pricing analysis. Capacity vs price. Energy rating vs price. Correlation analysis. Outlier analysis.

Phase 4 - Feature Engineering

One-hot encoding. Target encoding (optional). Derived features. Brand premium indicators. Specification complexity score.

Phase 5 - Model Development

Baseline: Linear Regression Advanced: Ridge Lasso Decision Tree Random Forest XGBoost Metrics: RMSE MAE R^2 Cross Validation

Phase 6 - Explainability

Feature importance. SHAP values. Price driver explanations.

Phase 7 - Pricing Intelligence Engine

User enters product specs and marketplace price. System predicts fair value. Compute: Premium Percentage Discount Percentage Value Score Verdict

Phase 8 - Alternative Finder

Find similar products. Rank by value score. Recommend better alternatives.

Phase 9 - Backend

FastAPI endpoints: /predict /analyze /alternatives /health

Phase 10 - Frontend

Pages: Home Analysis Dashboard Alternative Recommendations Model Explanation About Project

Deployment

Frontend: Vercel Backend: Render Model: Joblib serialization Database: PostgreSQL

Future Versions

V2: Live URL analysis. V3: Multi-site price comparison. V4: Personalized recommendations. V5: LLM-based product insights.

Interview Talking Points

Regression Feature Engineering Model Selection Bias-Variance Regularization Ensemble Learning Explainable AI Deployment End-to-End Product Thinking